that all electromagnetic waves—visible light among them—are the epitome of the peripatetic traveler. They never stop. They never slow down. Light always travels at light speed.

All is well and good until we ask, as the sixteen-year-old Einstein did, What happens if we chase after a beam of light, at light speed? Intuitive reasoning, rooted in Newton's laws of motion, tells us that we will catch up with the light waves and so they will appear stationary; light will stand still. But according to Maxwell's theory, and all reliable observations, there is simply no such thing as stationary light: no one has ever held a stationary clump of light in the palm of his or her hand. Hence the problem. Luckily, Einstein was unaware that many of the world's leading physicists were struggling with this question (and were heading down many a spurious path) and pondered the paradox of Maxwell and Newton largely in the pristine privacy of his own thoughts.

In this chapter we discuss how Einstein resolved the conflict through his special theory of relativity, and in so doing forever changed our conceptions of space and time. It is perhaps surprising that the essential concern of special relativity is to understand precisely how the world appears to individuals, often called "observers," who are moving relative to one another. At first, this might seem to be an intellectual exercise of minimal importance. Quite the contrary: In the hands of Einstein, with his imaginings of observers chasing after light beams, there are profound implications to grasping fully how even the most mundane situations appear to individuals in relative motion.

Intuition and Its Flaws

Common experience highlights certain ways in which observations by such individuals differ. Trees alongside a highway, for example, appear to be moving from the viewpoint of a driver but appear stationary to a hitchhiker sitting on a guardrail. Similarly, the dashboard of the automobile does not appear to be moving from the viewpoint of the driver (one hopes!), but like the rest of the car, it does appear to be moving from the viewpoint of the hitchhiker. These are such basic and intuitive properties of how the world works that we hardly take note of them.

Special relativity, however, proclaims that the differences in observations between two such individuals are more subtle and